Alvaro Gomariz

Imfeldstrasse 103, CH-8037 Zürich

+41 (0)76 773 14 72 | alvarogomarizc@gmail.com | alvarogomariz | 0000-0002-6172-5190 | Alvaro Gomariz

Zürich, 10th September 2026

Application for AI Scientist, Image Analysis & Digital Pathology (Req. REQ-10084320)

Dear Hiring Team,

I am excited to apply for this role because it sits at the intersection of the two things that have defined my work: extracting quantitative information from images, and connecting it to biological and clinical questions. Building AI for digital pathology to support oncology drug discovery is exactly where I would like to apply the combination of foundation-model, cell-level, and spatial imaging experience I have developed over the past decade.

At Roche, I led our early work in digital pathology. I technically guided a team developing a company-wide foundation model on tens of millions of H&E whole-slide images, adapting self-supervised architectures such as DINOv2 with distributed pre-training into a model that became an internal software package and three lead-author papers. In parallel, I built a proof-of-concept for lung-cancer patient stratification directly from H&E, using multiple-instance learning, graph neural networks, and survival analysis. Throughout, I worked closely with pathologists to ground the work in the clinical setting, interpret whole-slide images, and make model outputs interpretable.

This built on the foundation of my earlier research, where I developed deep learning for cell detection, phenotyping, and spatial analysis of multiplexed immunofluorescence and 3D microscopy, published in Nature Machine Intelligence, Science Advances, and Nature Communications. Quantifying cell types and their spatial relationships in tissue, across heterogeneous marker panels, is close to the questions this role asks of multiplexed imaging and spatial biology, and it is work I find genuinely exciting.

I also care about how this work is delivered: careful validation and benchmarking beyond accuracy, uncertainty-aware evaluation, and workflows and software that others can reproduce, maintain, and reuse. I am interested in how newer AI tooling, including foundation models and agentic approaches, can make scientific analysis and software development faster without compromising those standards.

More recently, I have led the technical and scientific direction of our AI video colonoscopy program: guiding a cross-functional team, owning the algorithm roadmap, and aligning scientists, engineers, biostatisticians, and clinicians on evidence that informs a late-stage clinical program. The imaging is different from pathology, but the program is deeply multimodal, combining endoscopic imaging with clinical data into prognostic models, and it has sharpened how I integrate heterogeneous data sources around a clinical question.

Across these projects I have worked at the interface of imaging, biology, and clinical development, sharing results through publications and conferences, and setting technical direction while aligning colleagues across functions. I would welcome the opportunity to bring this combination of digital pathology, foundation models, and quantitative cell and spatial imaging to the AI & Innovation team in Oncology Data Science. Thank you for considering my application.

Kind regards,

Alvaro Gomariz